

# The process stack

After the material stack and opening are built, the process stack transforms the profile step by step — the same idea as real fab steps (deposit, fill, etch, polish).

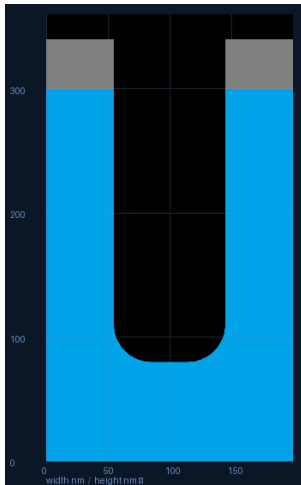
## How it works

Steps run **in order, top to bottom** (①, then ②, ③ ...). Each step acts on the result of the one above it. A **Repeat** block runs its steps  $\times N$  — handy for multilayers / superlattices.

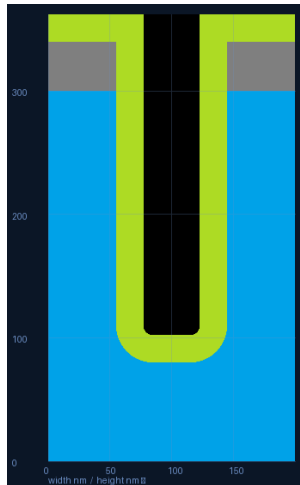
## The step types

- **Deposit · conformal** — coats every surface — walls, bottom, and top — with a uniform layer that follows the profile. Use it for liners and spacers.
- **Deposit · planar** — adds a flat layer on top only (no sidewall coverage).
- **Fill** — fills the opening/voids with a material (gap-fill), then continues above the surface.
- **Etch · isotropic** — removes material equally in all directions — rounds corners and undercuts.
- **Etch · anisotropic** — removes material vertically (directional). Anisotropy 1 = fully vertical, 0 = isotropic, in between = a rounded undercut. Pick a material to etch only that one, or “(any)”.
- **Planarize** — cuts everything flat at a height — like chemical-mechanical polish (CMP).

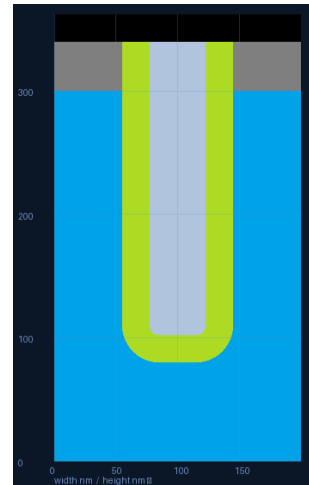
## Worked example — liner + fill + polish



① bare trench (stack + opening)



② + Deposit-conformal (nitride liner)



③ + Fill (tungsten) + Planarize

### Tips

- The material dropdown on Deposit/Fill/Etch lists only materials already in your stack (Etch adds “(any)”).
- Etch with a material selected removes only that material; “(any)” removes whatever it reaches.
- Build a superlattice with a Repeat block: e.g. deposit A, deposit B, repeated  $\times 10$ .